Firewall Configuration using iptables

Objective

To understand the basics of Linux firewall management using **iptables** by viewing existing firewall rules, creating rules to allow and block specific network traffic, verifying the rules, and finally removing them.

Tools Used

- Kali Linux
- iptables
- Terminal

Procedure

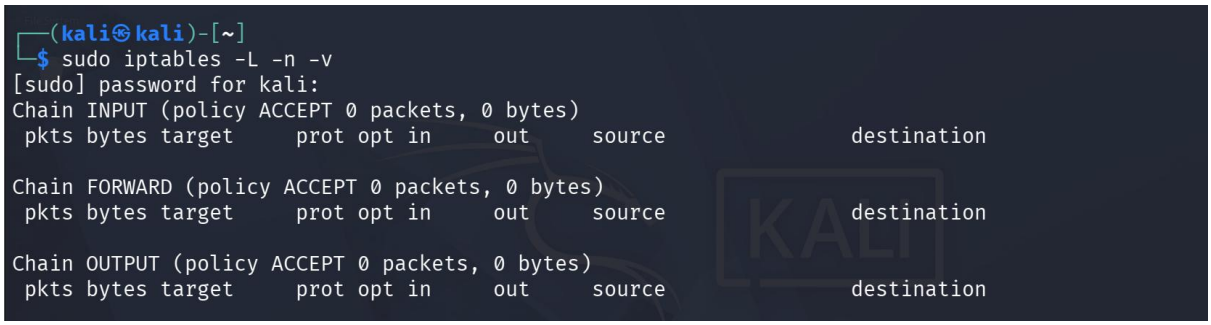
Step 1: Display Existing Firewall Rules

The following command was used to display the current firewall rules:

sudo iptables -L -n -v

Initially, no custom firewall rules were configured. All three default chains (**INPUT**, **FORWARD**, and **OUTPUT**) followed the **ACCEPT** policy.

Screenshot



Step 2: Allow SSH Traffic

A rule was added to allow incoming SSH connections on port **22**.

sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT

Step 3: Block Telnet Traffic

A rule was added to block incoming Telnet connections on port **23**.

sudo iptables -A INPUT -p tcp --dport 23 -j DROP

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Step 4: Verify the Firewall Rules

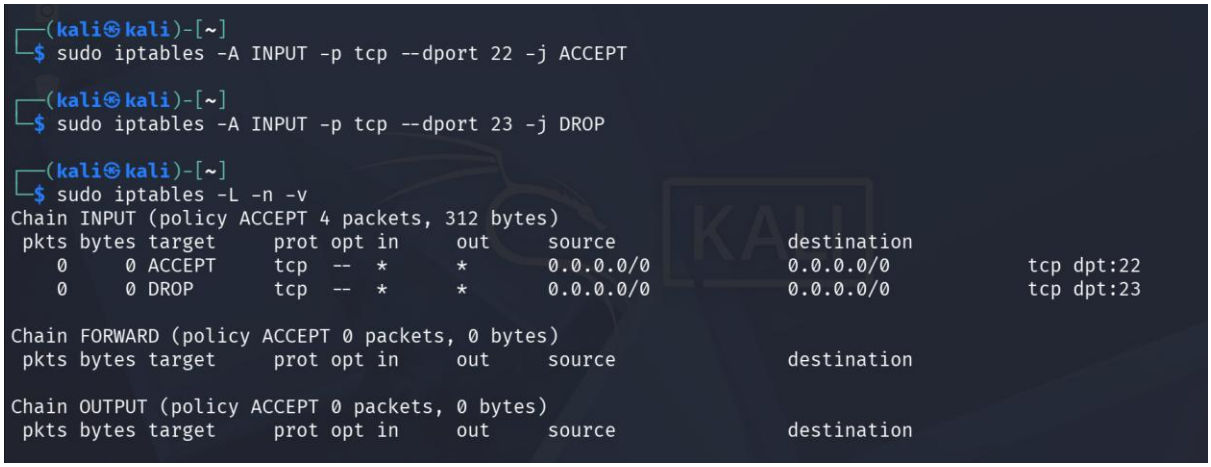
The configured rules were verified using:

sudo iptables -L -n -v

The output showed:

- TCP traffic on **port 22** is **ACCEPTED**.
- TCP traffic on **port 23** is **DROPPED**.

Screenshot



Step 5: Remove the Firewall Rules

After demonstrating the firewall configuration, all custom rules were removed using:

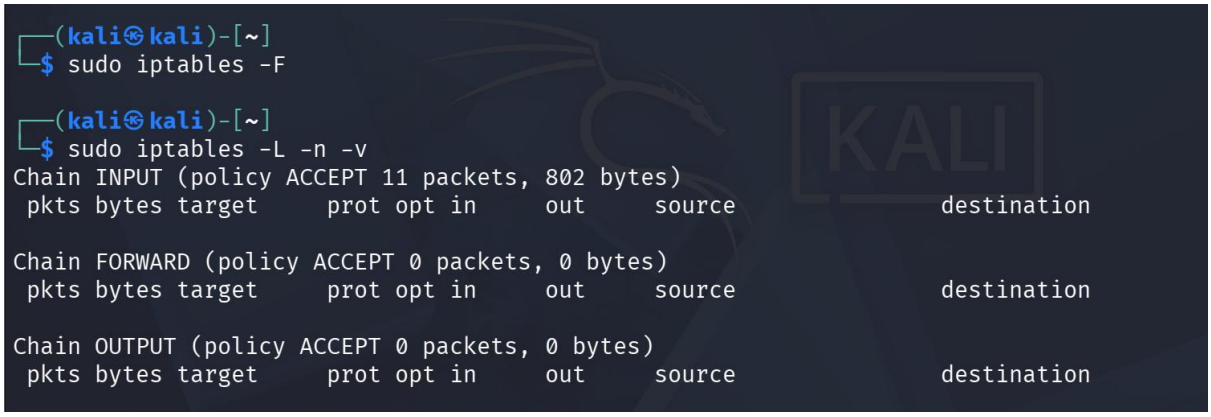
sudo iptables -F

The firewall table was checked again:

sudo iptables -L -n -v

The output confirmed that the custom rules had been successfully removed and only the default ACCEPT policies remained.

Screenshot



Observation

- Initially, no user-defined firewall rules were present.
- An **ACCEPT** rule was successfully created for SSH traffic (TCP Port 22).
- A **DROP** rule was successfully created for Telnet traffic (TCP Port 23).
- The configured rules were visible in the INPUT chain when listed.
- After flushing the firewall, all custom rules were removed successfully.

Conclusion

The experiment successfully demonstrated the basic functionality of **iptables** in Linux. Firewall rules were created to allow SSH traffic and block Telnet traffic, verified using the firewall rule list, and finally removed using the flush command. This exercise provided practical experience in configuring and managing firewall rules to control network traffic in a Linux environment.